

数字特征、大数定律-知识总结（仅数一三）

1.数字特征

1.期望:

$$EX = \sum_{k=1}^{+\infty} x_k \cdot p_k$$
$$EX = \int_{-\infty}^{+\infty} xf(x)dx; E[g(X)] = \int_{-\infty}^{+\infty} g(x)f(x)dx$$

2.方差:

$$DX = E(X^2) - E^2(X)$$
$$D(X \pm Y) = DX + DY \pm 2Cov(X, Y) = DX + DY \pm 2\rho\sqrt{DX}\sqrt{DY}$$

3.协方差:

$$Cov(X, Y) = E(XY) - EXEY$$
$$Cov(aX, bY) = abCov(X, Y)$$
$$Cov(X_1 + X_2, Y) = Cov(X_1, Y) + Cov(X_2, Y)$$

4.相关系数:

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{DX}\sqrt{DY}}$$
$$Y = aX + b \begin{cases} \rho_{XY} = 1, a > 0 \\ \rho_{XY} = -1, a < 0 \end{cases}$$
$$X \text{与} Y \text{不相关} \Leftrightarrow \begin{cases} \rho_{XY} = 0 \\ Cov(X, Y) = 0 \\ EXY = EX \cdot EY \end{cases}$$
$$X \text{与} Y \text{: 不相关} \not\Rightarrow \text{独立}, \text{独立} \Rightarrow \text{不相关}$$

5.常见分布:

分布	0-1	$B(n, p)$	$P(\lambda)$	$U(a, b)$	$E(\lambda)$	$N(\mu, \sigma^2)$
EX	p	np	λ	$\frac{a+b}{2}$	$\frac{1}{\lambda}$	μ
DX	$p(1-p)$	$np(1-p)$	λ	$\frac{(b-a)^2}{12}$	$\frac{1}{\lambda^2}$	σ^2

6.数字特征常用结论:

$$\begin{cases} E(\bar{X}) = EX, E(S^2) = \sigma^2, D(\bar{X}) = \frac{\sigma^2}{n} \\ X \sim \chi^2(n), EX = n, DX = 2n \end{cases}$$

2.期望:

1.计算分布

$$EX = \sum_{k=1}^{+\infty} x_k \cdot p_k; E[g(X)] = \sum_{k=1}^{+\infty} g(x_k) \cdot p_k$$
$$EX = \int_{-\infty}^{+\infty} xf(x)dx; E[g(X)] = \int_{-\infty}^{+\infty} g(x)f(x)dx$$
$$E[g(X, Y)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x, y)f(x, y)d\sigma$$

2.相互关系

$$EXY = EXEY \text{(当} X, Y \text{不相关时才成立)}$$
$$E(X^2) = DX + E^2(X)$$
$$E\bar{X} = EX$$
$$E(\chi^2(n)) = n$$

3.方差:

1.基本性质

$$D(a) = 0 \text{(其中} a \text{为常数)}$$
$$D(aX + b) = a^2 DX$$
$$D(aX \pm bY) = a^2 DX + b^2 DY \pm 2abCov(X, Y)$$
$$D(X \pm Y) = DX + DY \text{(要求} X, Y \text{不相关)}$$

2.初始计算

$$DX = E(X^2) - E^2(X)$$
$$D(\chi^2(n)) = 2n$$

4.大数定律

切比雪夫不等式:

$$P\{|X - EX| \geq \varepsilon\} \leq \frac{DX}{\varepsilon^2} \text{或} P\{|X - EX| < \varepsilon\} \geq 1 - \frac{DX}{\varepsilon^2}$$

大数定律

切比雪夫:

$$\begin{cases} \text{设} X_1, X_2, \dots, X_n, \dots \text{为两两不相关的随机变量序列, 存在常数} C \\ \text{使得} D(X_i) \leq C (i=1, 2, \dots), \text{则对于任意正数} \varepsilon, \text{有:} \\ \lim_{n \rightarrow \infty} P\left\{\left|\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n E(X_i)\right| < \varepsilon\right\} = 1 \end{cases}$$

伯努利:

$$\begin{cases} \text{设} X_1, X_2, \dots, X_n, \dots \text{相互独立, 同服从} 0-1 \text{分布, 则对任意正数} \varepsilon, \text{有:} \\ \lim_{n \rightarrow \infty} P\left\{\left|\frac{1}{n} \sum_{i=1}^n X_i - p\right| < \varepsilon\right\} = 1 \end{cases}$$

辛钦:

$$\begin{cases} \text{设} X_1, X_2, \dots, X_n, \dots \text{相互独立同分布, 且期望为} \mu, \text{则对于任意正数} \varepsilon, \text{有:} \\ \lim_{n \rightarrow \infty} P\left\{\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| < \varepsilon\right\} = 1 \end{cases}$$

中心极限定理

棣莫-弗拉普斯定理:

$$\text{设} X_i \sim B(n, p), \text{则对于任意实数} x$$
$$\text{有} \lim_{n \rightarrow \infty} P\left\{\frac{\sum_{i=1}^n X_i - np}{\sqrt{np(1-p)}} \leq x\right\} = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

列维-林德伯格定理:

$$\text{设} X_1, X_2, \dots, \text{相互独立且同分布, 且期望} \mu \text{方差} \sigma^2 \text{存在,}$$
$$\text{则对于任意实数} x, \text{有} \lim_{n \rightarrow \infty} P\left\{\frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} \leq x\right\} = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

依概率收敛:

$$\text{若} X_1, X_2, \dots, \text{是一个随机变量序列, } A \text{是一个常数, 若对于任意的} \varepsilon > 0,$$
$$\text{有} \lim_{n \rightarrow \infty} P\{|X_n - A| < \varepsilon\} = 1$$
$$\text{则称随机变量序列} X_1, X_2, \dots \text{依概率收敛于常数} A, \text{记作} X_n \xrightarrow{P} A$$